Time 2Hr **Operating System** MA41011/MA60004 Spring 2023

Instruction: Show complete rough work. No Calculator. No query.

1. A page on RAM has nuvyzst n:0 uvyzst wrong. n:1 Disk=RAM n:2 first variable correct on RAM. n:2{1} 3:{2} 4:{3} 5:{1,2} 6:{1,3} 7:{2,3} 8:{1,2,3}
   Initial Disk P:346728 Q:537946 R:823667 S:839258
   Initial RAM P:0296756 Q:0824589 R:0736318 S:0822839
   P:a,b,c Q:d,e,f R:g,h,i S:j,k,l print(d) makes Q:1537946 [537946 Disk→RAM]
   b=59 makes P:3295956 a=42 P:5425956 What does g=34; print(h); k=87; over do?

2. When job A..E are done. Example: P[896-1145] Q[0-896] R[896-1003] S[1145-1487]

Job	A	B	C	D	E		P	Q	R	S
Given	G	Y	H,M	K	Z		W	I,J	N	T
Will need	K,Z	M	U,L	H	Y		I	V	J	W,N
Service Time	42	34	27	80	17		249	896	107	342

3. Data compression block size 5: Complete0 is 00. Single one is 01<location in 2 bits>. Two one's 10<combined location in 4 bits>. More one's 11<block>
   101111 stands for single one at location 4. Hint:00011,10000 is coded as 101001,0100.
   Show coding of 01000,00001,01100,10101,

4. Let remaining service time of job P is 437. Let service time of Q is 390. Let swap out time is 42. Let swap in time is 35. When preemption is done find A) increase in waiting time of P B) decrease in waiting time of Q? See following example.

Job	Arrival	Service	Interval	Preemption	
U	0	30	0-30	0-30	RST(V)=40 ST(W)=13
V	20	50	30-80	30-40,65-105	Swap out time 7 Swap in time 5
W	40	13	80-93	47-60	Increase in waiting of V=35-10=25
					Decreasing in waiting of W=40-7=33

5. In following write missing numbers p, q, r, s. u, v Write name of scheme used.

Job	A	B	C	D	E	F		G
Memory needed	2000	222	879	303	1828	934	Job	283
Internal Fragmentation	42	124	0	134	p	r	B	u
External Fragmentation	0	0	2070	0	q	s	Over	v

6. Free space management in dynamic partitioning using best fit. Write contents of modified memory after each operation. A) Job U needs 8 memory B) Job V needs 2 memory C) Job Q which was allocated 45-49 is over. 99 is pointer to the first hole.

Address	08	25	42	55	67	80	99
Contents	25,9	55,3	67,5	80,6	8,8	nil,4	42,10
Free	00-08	23-25	38-42	50-55	60-67	77-80	89-98

7. Ready time of two jobs are 84 and 91 respectively. Their service times are 650 each. Find waiting in each. Write finish time also. Round robin time quantum 50.
   Example: RT(A):12 RT(B):27 ST(A):46 ST(B):38 Time quantum 20
   A:12-32 52-72 90-96 B:32-52 72-90 WT(A):20+18=38 WT(B):5+20=25

8. Directory entry for a file is k. Blocks are following. Write blocks when v is deleted.
   d:eqj- h:zs-y k:umch y:av-d Write all possible answers.

The end of Question Paper